PROJECT PORTFOLIO

# Redesigning the Healthy City Dashboard – Mode Share Indicator Page

ASSIGNMENT 1.7

Presented by:
Char Alimot, Hilal Malik

# Project Overview

The Healthy City Dashboard — Mode Share Indicator Redesign is a 12-week collaborative project focused on improving how Vancouver residents, planners, and advocacy groups consume open transportation data (biking, walking, transit use).

The original Mode Share page had major usability barriers:

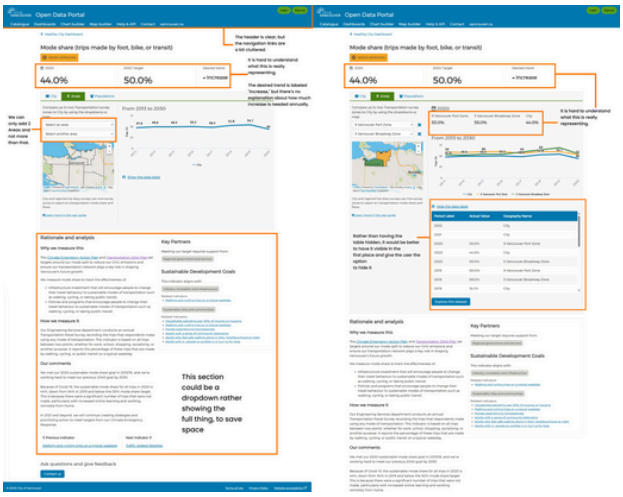
- Static tables
- Low visual hierarchy
- Limited filtering
- Inconsistent accessibility

Our project aimed to redesign the page to be clearer, more intuitive, more accessible, and more engaging, while remaining aligned with the City of Vancouver's existing Healthy City Dashboard ecosystem.

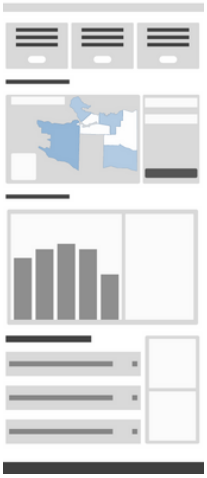
**Project Outcomes**

- Full UX redesign (lo-fi → hi-fi)
- Data-driven interface coded with HTML/CSS/JS + JSON
- Interactive charts (Chart.js) with tooltips, labels, and filters
- A stable desktop-first layout
- End-to-end PM artifacts (roadmap, WBS, risks, QA, RACI, MoSCoW)

Final integrated, polished dashboard ready for presentation




**Content Audit**



**Lo-fi**



**Hi-fi**

# Team Roles & Responsibilities

**Char – Project Manager, Design Lead, Supporting Developer**

- Led project planning and documentation (scope, roadmap, WBS, risks, RACI, MoSCoW).
- Owned UX/UI design for lo-fi and hi-fi prototypes, including layouts, flows, and visual hierarchy.
- Built the main HTML/CSS layout so the desktop dashboard matched the design system.
- Guided accessibility and interface copy, and tested the integrated build to flag issues and help polish the final dashboard and presentation.

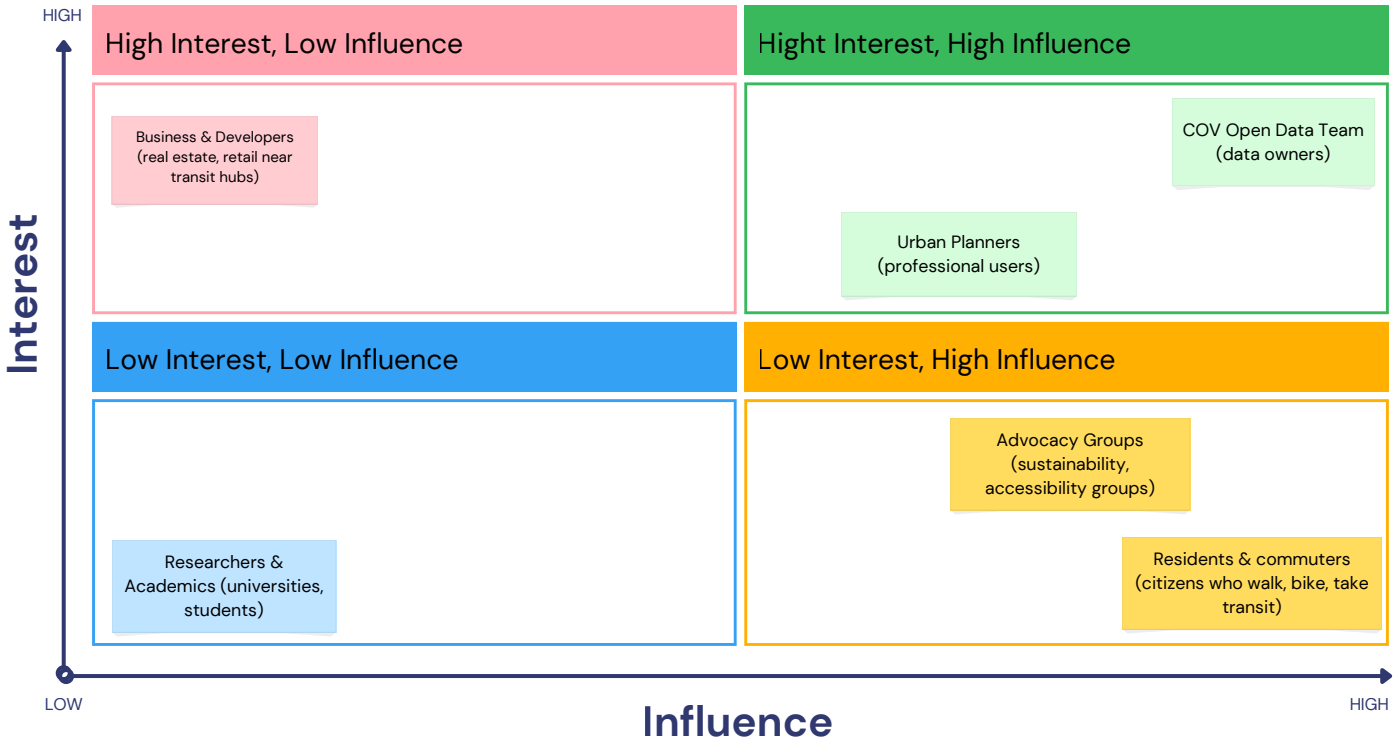
**Hilal – Development Lead, Supporting Designer**

- Led front-end development and JSON data integration for the Mode Share dashboard.
- Implemented interactive charts, map behaviour, and filtering logic.
- Handled JavaScript, data accuracy checks, debugging, and performance.
- Supported design decisions by giving technical feedback on layout and interactions.

# RACI Matrix

| Deliverable | PM (Char) | Design Lead (Char) | Dev (Hilal) | Instructors (Joanna & Gill) |
|---|---|---|---|---|
| **Project Setup** (workspace) | A | C | R | I |
| **Lo-fi Prototype** (research, wireframes, personas, brand identity) | A | R | C | C |
| **Hi-fi Prototype** (hi-fi screens, design system, interactivity, peer testing) | A | R | C | C |
| **Functional Code** (JSON data, charts/tables, accessibility, navigation) | A | C | R | C |
| **Final Dashboard & Presentation** (integration, polish, presentation prep) | A | R | R | C |

# Stakeholder Map




# Purpose, Scope & Goals

**Purpose**

The purpose of this project was to redesign the Mode Share page so users can interpret transportation data more clearly, interactively, and confidently.

**Scope**

The project focused on auditing the existing Mode Share page, redesigning it from lo-fi to hi-fi, implementing a JSON-driven coded version, integrating design and code into a cohesive dashboard, and running QA to prepare for a reliable live demo.

**Core Goals**

Our goals were to make transportation data easier to read and explore, improve navigation and accessibility, support both everyday residents and planners, build trust through clear data context, and deliver an engaging, desktop-optimized experience with interactive charts.

# Roadmap Weeks 1-12

**HCDR-20 – Project Start (Week 1: Sept 3)**
Establish project objectives, define roles, confirm scope, and align expectations with stakeholders.

**HCDR-23 – Research & Low-Fidelity Concept (Weeks 2–5: Sept 9 – Oct 6)**
Conduct usability analysis of the existing dashboard, identify opportunities for improvement, and create low-fidelity wireframes to test layout and flow.

**HCDR-32 – Hi-Fi Prototype (Weeks 6–7: Oct 7 – Oct 21)**
Develop a high-fidelity interactive prototype with refined visuals, colour systems, and navigation flow based on feedback from users and instructors.

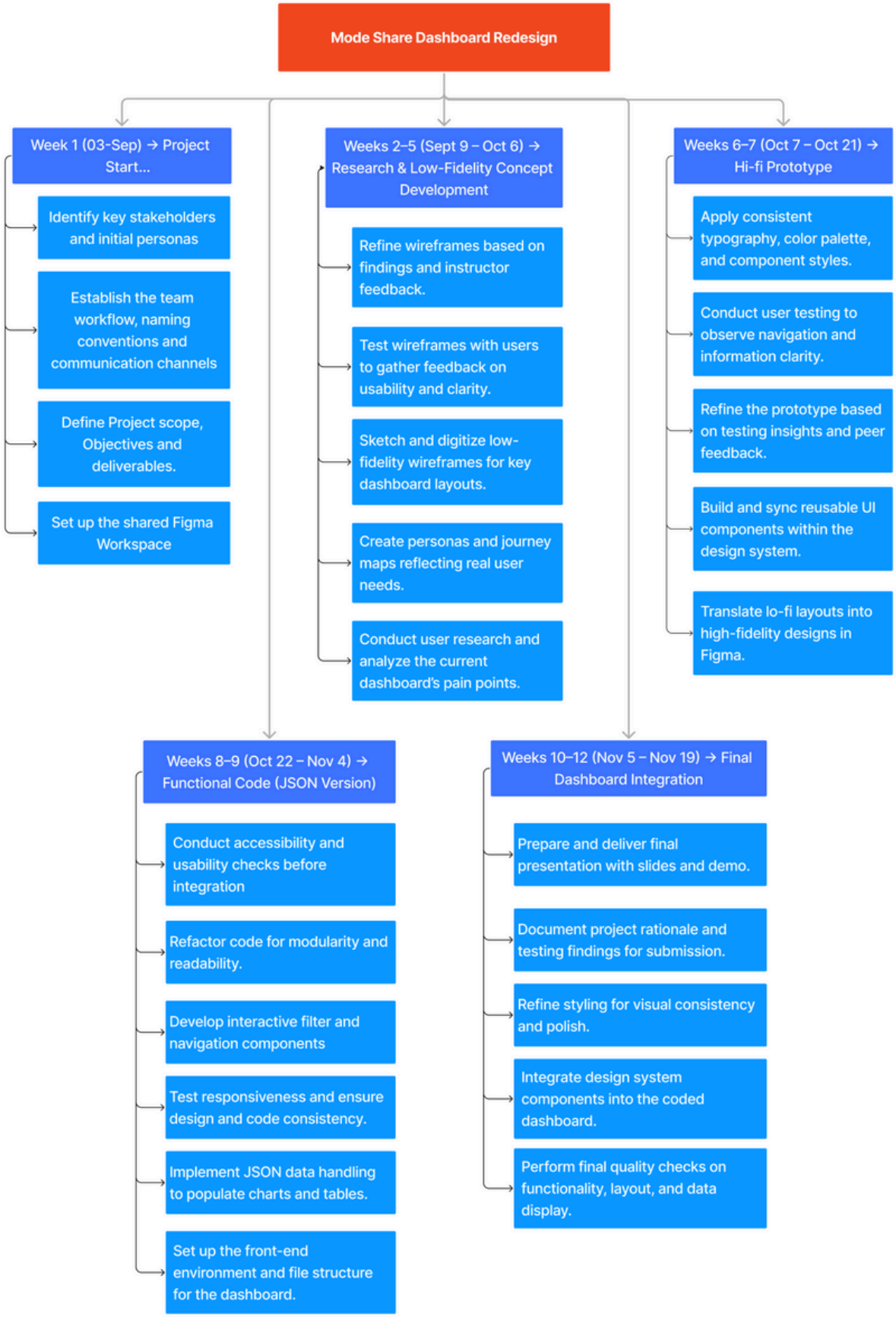
**HCDR-43 – Functional Code (JSON Version) (Weeks 8–9: Oct 22 – Nov 4)**
Implement core front-end functionality and connect static JSON data to charts using Chart.js, ensuring accurate data display and interactivity.

**HCDR-54 – Final Dashboard Integration (Weeks 10–12: Nov 5 – Nov 19)**
Combine design and code into a complete, accessible, and responsive dashboard. Conduct usability testing, finalize documentation, and prepare for the final presentation.

# Gantt Chart



# WSB


Mode Share Dashboard Redesign
Week 1 (03-Sep) → Project Start...
Identify key stakeholders and initial personas
Establish the team workflow, naming conventions and communication channels
Define Project scope, Objectives and deliverables.
Set up the shared Figma Workspace
Weeks 2–5 (Sept 9 – Oct 6) → Research & Low-Fidelity Concept Development
Refine wireframes based on findings and instructor feedback.
Test wireframes with users to gather feedback on usability and clarity.
Sketch and digitize low-fidelity wireframes for key dashboard layouts.
Create personas and journey maps reflecting real user needs.
Conduct user research and analyze the current dashboard's pain points.
Weeks 6–7 (Oct 7 – Oct 21) → Hi-fi Prototype
Apply consistent typography, color palette, and component styles.
Conduct user testing to observe navigation and information clarity.
Refine the prototype based on testing insights and peer feedback.
Build and sync reusable UI components within the design system.
Translate lo-fi layouts into high-fidelity designs in Figma.
Weeks 8–9 (Oct 22 – Nov 4) → Functional Code (JSON Version)
Conduct accessibility and usability checks before integration
Refactor code for modularity and readability.
Develop interactive filter and navigation components
Test responsiveness and ensure design and code consistency.
Implement JSON data handling to populate charts and tables.
Set up the front-end environment and file structure for the dashboard.
Weeks 10–12 (Nov 5 – Nov 19) → Final Dashboard Integration
Prepare and deliver final presentation with slides and demo.
Document project rationale and testing findings for submission.
Refine styling for visual consistency and polish.
Integrate design system components into the coded dashboard.
Perform final quality checks on functionality, layout, and data display.

# Moscow

| Priority | Feature / Requirement | Rationale |
|---|---|---|
| Must Have | Interactive mode share chart (car, transit, walking, cycling) | Core to understanding transportation data |
| Must Have | Ability to filter or select specific regions or years | Essential for comparing data |
| Should Have | Clear tooltips and legends explaining terms | Improves comprehension for non-technical users |
| Should Have | Data comparison view (side-by-side or overlay) | Allows users to compare areas or modes directly, supporting deeper insight and analysis |
| Could Have | Download or share data button | Enhances usability for researchers and student |
| Could Have | Animated transitions between data states | Adds engagement and visual polish |
| Won't Have | Live data updates from city API | Out of scope for prototype but a possible future feature |

# Execution Phase

**Lo-Fi Prototype**
We audited the existing dashboard, reviewed content, developed personas and scenarios, and explored multiple low-fidelity layout options. We also mapped key workflows to understand how users move through the Mode Share page.
Outcome: a validated layout structure and a clear design direction shaped by feedback.

**Hi-Fi Prototype**
The approved lo-fi wireframes were turned into a polished high-fidelity prototype and design system, including colour palette, typography, chart explorations, and reusable components such as cards, labels, and filters.
Outcome: a refined, accessible interface that became the foundation for the coded dashboard.

# Execution Phase

Risk Mitigation Plan Status Update:

- Time Management and Scheduling
  - There is a risk that competing coursework and deadlines may lead to insufficient time for development and testing, resulting in incomplete features or lower project quality.
  - Mitigation: Plan deliverables week by week, set internal deadlines ahead of official due dates, and prioritize core functionality first.
    - Status update: Gil extended the deadline by 2 days due to the Figma outage, but we still finished on time following our tight plan. We used the extra time for self-testing. Week 8 studio time was also used for peer testing/feedback, which informed final Hi-Fi tweaks.

- Technical Complexity
  - There is a risk that the chosen charting or data visualization library may lead to technical difficulties or implementation delays, resulting in reduced interactivity or reliance on static visuals.
  - Mitigation: Prototype with the library early, and if challenges persist, switch to a simpler or more stable option (e.g., Chart.js).
    - Status update: We followed the mitigation plan and started chart visuals early, giving us time to test and fix animations/overlays.

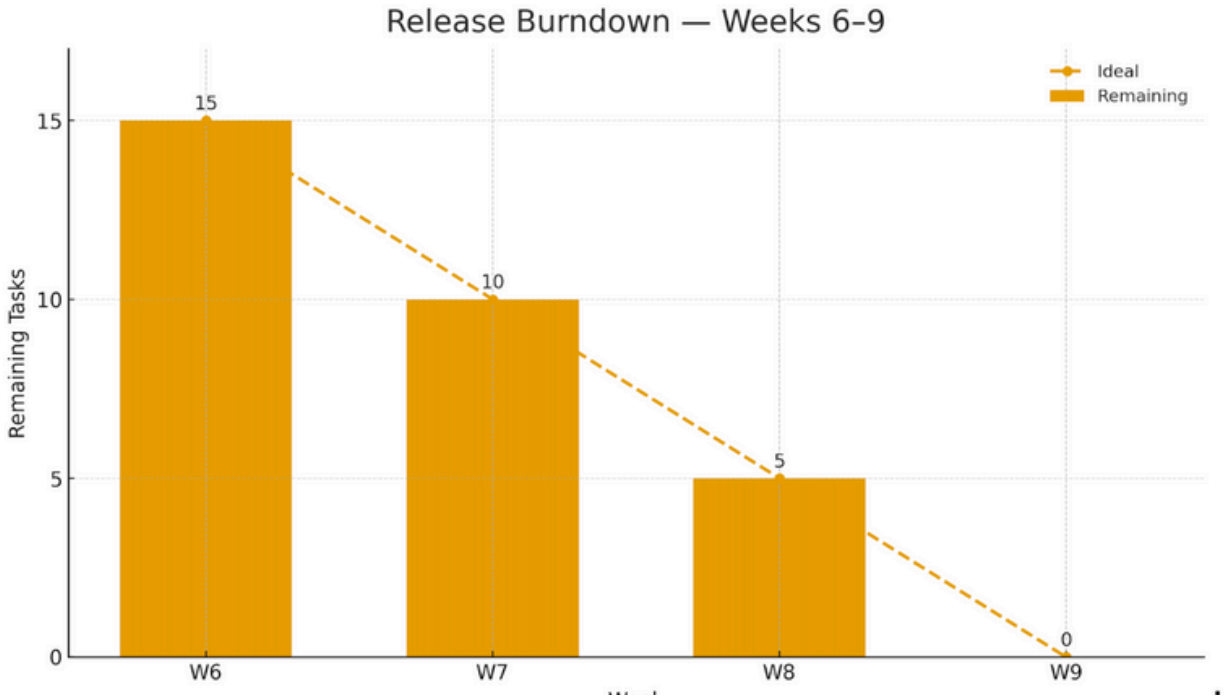
- Communication and Coordination
  - There is a risk that team members missing check-ins or unclear role distribution may lead to coordination issues and uneven workload, resulting in delays or inconsistent outputs.
  - Mitigation: Define clear roles (Design Lead, Dev Lead), maintain weekly progress updates, and track contributions using shared documentation.
    - Status Update: Roles are confirmed. Daily status and tracking through Jira, Slack, and Discord keeps everything visible and accounted for.

- Data Integration Issues
  - There is a risk that errors in linking JSON data to the front-end interface may lead to inaccurate or missing data displays, resulting in loss of dashboard credibility and usability.
  - Mitigation: Test data connections early with small samples, validate outputs frequently, and document integration steps clearly.
    - Status Update: Still testing charts and wiring them to the JSON data.

# Execution Phase

- Accessibility Oversight
  - There is a risk that accessibility standards (contrast, labelling, alt text) may be overlooked during design and coding, resulting in a non-inclusive user experience.
  - Mitigation: Assign one team member to focus on accessibility compliance (WCAG 2.1 AA) and review designs before implementation.
    - Status Update: Prioritizing functional charts + JSON first; accessibility code (contrast, labels, alt text, keyboard) follows immediately after integration.

**Coding Phase (A03 – JSON Functional Code)**
We then built the functional dashboard by integrating JSON data, adding interactive filters, and implementing chart components. The team ensured data accuracy, improved accessibility, refined responsiveness, and kept the codebase modular.
Outcome: a working data-driven dashboard ready for final integration and visual polish.




From Weeks 6–9, the team handled 15 tasks. Following Gil's guidance, they prioritized the JSON chart over the GeoJSON map to avoid delays. With this narrower scope, they completed the JSON functional code within the weekly sprint and stayed on schedule, by Week 9, all work was done. Char kept the team on a strict schedule, so there were no late tasks. The weekly sprints also made it easier to split deliverables and kept the workload manageable for a team of 2.

# Execution Phase

**Final Integrated Code (A04 – Integrated Dashboard)**

In the final phase, we focused on bringing everything together into one cohesive desktop dashboard. We integrated the functional JSON-based charts with the hi-fi design, applied the design system consistently across components (colour, typography, spacing), and refined the layout so key sections lined up cleanly at common desktop widths. We also tightened navigation labels and active states, smoothed out the main demo flow, and ran multiple rounds of QA to check data accuracy, readability, and overall behaviour.

Outcome: a polished, stable integrated dashboard that matched the original design direction, supported a smooth live demo, and was ready to present to stakeholders as a finished product.

# Integration Phase

**Testing & Issue Log**

During integration, the team focused on identifying and resolving issues that affected polish, clarity, and overall consistency. We found inconsistencies in how the design system was applied across screens, chart font size and contrast problems that made some labels harder to read, misalignment at certain desktop breakpoints, and a few outdated or unclear navigation labels. We documented these issues systematically, grouped them by impact, and prioritized which ones to address first. Working through these fixes, especially around spacing, typography, and labelling, made the dashboard feel more cohesive, professional, and stable for the final demo.

**QA Findings**

After multiple QA passes, all main user flows, opening the dashboard, exploring data by mode or year, and comparing areas, worked reliably end to end. No critical bugs remained, and the final tweaks were limited to minor spacing changes, wording updates, and small visual refinements. The desktop demo became smooth and predictable, with charts loading correctly and navigation behaving as expected. This gave the team confidence that stakeholders would be able to follow the story without being distracted by glitches or inconsistencies during the final presentation.

# Integration Phase

**Retrospective Summary**
Looking back, unifying the design system had a noticeable impact on how "finished" the product felt, and practicing the demo flow several times made the narrative much clearer and easier to deliver. Visual alignment changes, such as tightening spacing and standardizing card layouts, also made the data easier to scan and compare at a glance. At the same time, integration took longer than expected, and the team had to let go of some microcopy tweaks and extra polish in order to protect the core experience and meet the deadline.

From this, the team learned the value of locking features earlier, giving integration and QA more breathing room, testing the demo path as if we were real stakeholders seeing the dashboard for the first time, and keeping explanations simple so the focus stays on insights and decision-making, not just on the interface itself.

# Team Reflection

Looking back on the project, we're proud of how we combined our strengths to deliver a clear, functional redesign of the Mode Share dashboard. Collaborating across design, development, and project management helped us stay aligned, even when timelines were tight or tasks became more complex. Moving through each milestone from research and low-fi sketches to JSON integration and final QA deepened our understanding of how design decisions and technical constraints affect each other.

Along the way, we learned the value of communicating early, validating ideas through testing and feedback, and relying on a consistent design system to make integration smoother. The project also pushed us to balance ambition with feasibility, prioritizing what had the most impact for users instead of trying to ship every enhancement. Overall, this experience strengthened our ability to work as a team, iterate thoughtfully, and deliver a complete product from concept to final code, with design and development perspectives fully integrated.